

## Goal for Today's Meeting:

Decide on our analysis plan for the research proposal and assign tasks to move the project forward.

## What We Did:

- Re-ran the DADA2 denoising code on our dataset with the truncation length set to 252 instead of 253, and uploaded the QIIME2 files containing the new truncation length to GitHub
- Filtered out the mitochondria/chloroplast genes and taxa without sufficient taxonomic information from the data (i.e. taxa without a designated phylum)
- Structured and organized the GitHub repository to ensure it's ready for upcoming data analysis work

*Note: We did not re-run the rarefaction with the sampling depth set to ~8000 instead of 3145 since we will be using the new value suggested in RStudio for analysis of diversity metrics*

## What We Plan to Do:

### Aim 1 - Diversity Analysis

- Initially run various diversity metrics to see how they affect the data
- Generate curves for all of the alpha-diversity metrics (e.g. Observed Features, Shannon Index, Simpson Index, Chao1 Index, and Faith's Phylogenetic Diversity)
- Run and generate curves for all beta-diversity metrics as well, but only select the most interpretable metric for the final analysis
- Finalize the statistical approaches (likely PERMANOVA) we will use to test for significance and size of effects

*Note: all curves using various alpha- and beta-diversity metrics will be coloured by histological cancer stage and likely shaped by sex*

### Aim 2 - Taxonomic Differences

- Generate the relative abundance summaries with at least the genus level and potentially the species level if there is sufficient information in the sample reads
- Compare and contrast the relative abundance of the various taxonomies across the histological stages
- If beta diversity differs across stages and/or with sex stratification, determine which taxa contribute most to the observed separation

### Aim 3 - Functional Analysis

- Perform an initial exploratory analysis of all functional pathways, rank them by magnitude of change across stages, and shortlist the top ~20 pathways to assess whether common biological themes, such as metabolism, emerge.
- Use existing literature to help guide selection of relevant pathways (e.g., metabolic, inflammatory, tissue repair processes associated with gastric cancer), then test and visualize whether these pathways differ across stages and/or by sex.
- Create a list of carcinogenic bacteria through the literature and see if any appear in our statistical analyses/work

## Timeline + Task Assignment:

- Assign responsibilities and expectations for Group Assignment 2
- Possibly set internal deadlines for the research proposal
- Look into any literature that used the same dataset and reported findings relevant to our research question